VF Asia Footwear Laboratory Testing Manual

Test Method for pH of an Aqueous Extract of a Material	Test method # VF- E20 REF # : TM-R- 061 Page 1 of 5
Date Issued : January 17, 2017 Revision # : 2 Reason for revision: Added option II - ISO 3071: 2005 and Option III- ISO 4045 : 2008	Created by : Nila Babia Erazo

PURPOSE AND SCOPE: To determine the acidity /alkalinity of an aqueous extract of a material such as all types of leather, leather-boards, fiberboards, and textiles.

REFERENCES: ASTM D 2810, SATRA TM 349, ISO 3071: 2005, ISO 4045:2008

EQUIPMENT AND MATERIAL:

1. A cutting device:
   - A pair of sharp scissors
   - Mill with stationary and rotating knives, speed of rotation 850 ± 150 rev/min*.(Optional)*
2. Strainer (sintered glass or non- absorbent nylon cloth) or Centrifuge Machine
3. Electrode / A device for measuring pH in the range of 0 – 14 to the nearest 0.01 pH unit.
4. Thermometer, capable of measuring to 0.5 ° C.
5. A laboratory balance capable of measuring mass up to 50 g to nearest 1 mg.
6. A device for measuring volumes up to 100 cm^3 to the nearest 1 cm^3. A measuring cylinder or volumetric flask is suitable.
7. Glass-wares :
   - Flasks, plastic or borosilicate glass, with a close fitting stopper. Soda glass flasks are unsuitable as the material may affect the pH value.
   - Beakers , chemically resistant
8. Glass stirrer, rods which is chemically resistant.
9. Mechanical shaker either of the following types *(Optional)*
   - Capable of holding each of the flasks and inverting them 100 ± 20 times per minute
   - Provides rotational or reciprocating movement
10. Water bath or hotplate
11. Tissue paper or white clean cloth.
12. leather, leather-boards, fiberboards, and textiles
13. Reagents :
    - Distilled or deionized water
      - Free from carbon dioxide and ammonia
      - With a pH of between 6.00 and 7.00
      - With a maximum conductivity of 1mS/m

    NOTE: Distilled water should at least grade 3 as defined in ISO 3696. If the distilled water is not grade 3, boil 100 ml of distilled water in a beaker at a moderate rate for (10 ±1) min and allow the covered beaker to cool to room temperature. VF Test method recommends a pH value between 6.0 and 7.0.
    - Potassium chloride solution, 0.1 mol. / l, prepared using distilled or deionized water.
    - pH buffer solutions , of pH 4 , 7, and 10

Test Method for pH of an Aqueous Extract of a Material	Test method # VF- E20 REF # : TM-R- 061 Page 2 of 5
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SAMPLE PREPARATION:

1. Material should be conditioned 23 ± 2°C and 55± 10% RH for a period of 48 hours prior to testing.
2. Cut the laboratory sample into pieces having approximately 2. 5 mm x 2.5 mm or of such size to allow the test samples to wet out rapidly.
   Note : If the test material is wet or damp, dry it in a warm environment of temperature less than 50°C and then condition for 48 hours at standard laboratory environment. Do not cut wet or damp material in the mill.
3. To avoid contamination, handle the material as little as possible.

REPETITIONS: Test three (3) samples per material

TEST PROCEDURE:

Calibration of pH Meter:

1. Standardize the pH meter with two buffer solutions; one below the expected value and one above the expected value. Normally pH 4 and pH 7 is used.
2. In case a pH of 7 is obtained, repeat the test and use buffer solution pH 7 and pH10. Both these buffer readings shall be within 0.02pH unit of the correct reading when the meter is standardized.
   - for acid range: use pH 4.0 and pH 7
   - for alkaline range: use pH 7 and pH 10

   Note: Used buffer solution should be discarded
3. Calibrate the pH meter to the temperature of the extract solution at 20±2°C.
4. The value should be near to the pH of the buffer solution.
5. Rinse the electrode with distilled water and blot dry.
6. After the test, always keep the electrode wet with KCL or distilled water. Formation of salt during standing does not affect the test.

Option I: Test Method ASTM D 2810, SATRA TM 349,

1. Use the balance to measure the mass of the test material. The mass should greater than 3 g but less than or equal to 5 g. Record this mass as [Mo].
2. Calculate the volume of water [Vo] required, in cm^3 to the nearest cm^3, by using the formula:
   [Vo] = [Mo] x 100
3. Use the thermometer to measure the temperature of the water. If it is outside the range of 18 to 22°C warm or cool it accordingly.

Test Method for pH of an Aqueous Extract of a Material	Test method # VF- E20 REF # : TM-R- 061 Page 3 of 5
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4. Place the test specimen in a flask and add the volume of water.
5. Stopper the flask and shake it vigorously by hand for 30 ± 5 seconds.
6. Let the flask stand at the standard laboratory conditions of 23 ± 2°C and 55 ± 10% RH, with occasional agitation for a period of 6.0 ± 0.2 hours. ***Note: If a shaking machine is available load the flask and operate the shaking machine providing rotational or reciprocating movement at 100 ± 20 times per minute and the water is maintained at a temperature of 23 ± 2°C for a period of 6.0 ± 0.2 hours.***
7. Allow the extract to settle before decanting. If difficulty is experienced in decanting the extract from the slurry, it may be strained through a clean, dry, non- absorbent mesh (for example, nylon cloth or coarse sintered glass filter), or centrifuged.
8. Calibrate the electrode of the pH meter. Refer to procedure on “Calibration of pH Meter.
9. Fully immerse the electrode of the pH meter in the flask with the aqueous extract
10. Once the reading on the pH meter has stabilized, record the reading to the nearest o.1 pH units.

Option II: Test Method ISO 3071:2005 for Fabric

1. Prepare the test sample in triplicate at room temperature.
2. Place each test sample of 2.0 ± 0.05 g each and 100 ml of extracting solution into a stoppered flasks. Refer to reagent description.
3. Agitate the flask by hand for at least 30 seconds to ensure that the textile material is properly wetted out, then shake it mechanically. ***The shaking machine should provide rotational or reciprocating movement at 100 ± 20 times per minute and the water is maintained at a temperature of 23 ± 2°C for a period of 2.0 hours ± 5 minutes.*** Record the temperature of the extracting solution used.
4. In the absence of mechanical shaker, agitate or stir the diluent in the beaker using a glass rod till uniformly wetted and let stand overnight with cover.
5. Allow the extract to settle before decanting. If difficulty is experienced in decanting the extract from the slurry, it may be strained through a clean, dry, non- absorbent mesh (for example, nylon cloth or coarse sintered glass filter), or centrifuged.
6. Calibrate the electrode of the pH meter. Refer to procedure on “Calibration of pH Meter”.
7. Decant the first extract into a beaker, immediately immerse the electrode to a depth to at least 10 mm and stir gently with a rod until the indicated pH-value stabilizes (do not record the pH-value of this solution).
8. Decant the second extract into another beaker, immediately immerse the electrode, without washing, in the beaker to a depth of at least 10 mm and allow to stand without stirring until the pH –value stabilizes. Record this value.
9. Decant the second extract into another beaker, immediately immerse the electrode, without washing, in the beaker to a depth of at least 10 mm and allow to stand without stirring until the pH –value stabilizes. Record this value.
10. Record the pH-values of the second and third extracts as the first and second measurements.

Test Method for pH of an Aqueous Extract of a Material	Test method # VF- E20 REF # : TM-R- 061 Page 4 of 5
Date Issued : January 17, 2017 Revision # : 2 Reason for revision: Added option II - ISO 3071: 2005 and Option III- ISO 4045 : 2008	Created by : Nila Babia Erazo

Note: If the difference between the two pH-values, expressed to the nearest 0.1 pH-units, is greater than 0.2, repeat the procedure with other test samples. When two valid measurements have been obtained, calculate the mean value.

Option III: Test Method ISO 4045:2008 for Leather

1. Prepare the extract in triplicate at room temperature.
2. Place each test sample of 5.0 ± 0.05 g each and 100 ml of extracting solution into a stoppered flasks. Refer to reagent description
3. Agitate the flask by hand for at least 30 seconds to ensure that the textile material is properly wetted out, then shake it mechanically. ***The shaking machine should provide rotational or reciprocating movement at 100 ± 20 times per minute and the water is maintained at a temperature of 23 ± 2°C for a period of 6.0 hours ± 5 minutes.*** Record the temperature of the extracting solution used.
4. In the absence of mechanical shaker, agitate or stir the diluent in the beaker using a glass rod till uniformly wetted and let stand overnight with cover.
5. Allow the extract to settle before decanting. If difficulty is experienced in decanting the extract from the slurry, it may be strained through a clean, dry, non- absorbent mesh (for example, nylon cloth or coarse sintered glass filter), or centrifuged.
6. Calibrate the electrode of the pH meter. Refer to procedure on “Calibration of pH Meter”.
7. Immediately, dip the electrode to the diluent to a depth of at least 10mm and stir with a rod gently until a steady reading is reached. The reading shall be taken within 30"~60" after rinsing the electrodes in the extract, [pH1].
8. If the pH value is below 4 or over 10, the difference figure shall be determined.

- Using a pipette, transfer 10ml of extract into the volumetric flask and make up the mark with water (100ml).
- Rinse the electrodes with approximately 20ml of the diluted solution and then measure the pH value as in (step # 7), [pH2].
- Calculation : The difference is calculated as follows:
  pH = [pH2] - [pH1]

NOTE : Interpretation of result if pH is below 4.0 and above 10.

1. pH is between 0.00 and 3.95.

a) If the acid present in the extract is a strong acid the pH will change by more than 0.70 pH units, due to the high dissociation of the acid. A change in pH of more than 0.70 after dilution is an indication of the presence of strong acids.

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b) If the acid present in the extract is a weak acid the pH will change by less than 0.70 pH units, due to the low dissociation of the acid.

2. pH is between 10.05 and 14.00

   a) Determine the difference figure in the same way as in # 1 in order to look for strong bases and apply the same criteria of 0.70 pH change.

3. Difference figure result:
   a) Difference figure < 0.70, **PASS.**
   b) Difference figure > **=** 0.70,**FAIL**

REPORT:

1. The mean value of the two readings to the nearest 0.1 pH units.
2. Include :
   - Material tested
   - Any deviations from this procedure
   - pH of extracting solution or distilled water.